

AI453 — Probabilistic Graphical Models

Practical #1 — Introduction to Probability Theory

Department of Artificial Intelligence, SVNIT Surat · B.Tech. AI, Semester VII

Read this first

Nothing to install. Plain Python 3 — no NumPy, no pip, no virtual environment. If Python runs on your machine, you are ready. If it does not, use Google Colab.

There is no dataset. You are given one joint distribution, printed below. Every number you produce today comes out of that one table. Nothing is loaded from a file, downloaded, or estimated from samples. *The distribution is the input*, and the work is manipulating it.

Setup

Download `lab1_simple.py`, then:

```
python3 lab1_simple.py
```

It should print one line: $P(A=1) = 0.5$. That is the worked example. Everything else in the file is yours to write.

Learning goal

Compute marginals, joints, conditionals, the chain rule and Bayes' rule directly from a joint distribution — everything from Lecture 1, done first by hand and then by machine.

The distribution you are given

$P(A, B, C)$ — three binary variables, eight numbers, summing to 1

A	B	C	$P(A, B, C)$
0	0	0	0.06
0	0	1	0.24
0	1	0	0.04
0	1	1	0.16
1	0	0	0.09
1	0	1	0.06
1	1	0	0.21
1	1	1	0.14

In the starter file this is a Python dictionary. The key (a, b, c) means “ $A=a$ and $B=b$ and $C=c$ ”, so $P[(1, 0, 1)]$ is $P(A=1, B=0, C=1) = 0.06$.

The only pattern you need

The starter file contains one worked example — $P(A=1)$:

```
total = 0.0
for (a, b, c), p in P.items():
    if a == 1:
        total += p
print("P(A=1) =", total)
```

In words: *walk through all eight rows, and add up the ones where A is 1.*

That is the whole idea

A **marginal** is a sum over the rows that match. A **joint** is the same sum with a longer condition. A **conditional** is one such sum divided by another. Every task below is that loop with a different **if**.

Your tasks (T1–T8)

How to work through these

Do T1–T6 on paper first. Each one is adding up a handful of numbers from the table — ten seconds with a pen. *Then* write the loop and check that the computer agrees with you. If it does not, one of you is wrong, and finding out which is the real exercise.

T7 is the one where the loop earns its keep: checking the chain rule by hand means working out three separate quantities for each of the eight rows. Let the computer do that one.

- T1.** Check that the table is a valid distribution: all eight numbers must add up to 1. Print the total.
- T2.** Compute and print $P(B=1)$. Same loop as the worked example, different condition.
- T3.** Compute and print $P(C=1)$.
- T4.** Compute and print the joint $P(A=1, B=1)$. Now the condition has two parts.
- T5.** Compute and print the conditional $P(C=1 \mid A=1)$:

$$P(C=1 \mid A=1) = \frac{P(A=1, C=1)}{P(A=1)}$$

Two sums, one divided by the other. You may compute the top and the bottom in the same loop.

- T6.** Compute and print $P(B=1 \mid A=0, C=1)$. Two things known, one thing asked — same pattern.
- T7. The chain rule.** In class we derived, for any three variables,

$$P(A, B, C) = P(A) P(B \mid A) P(C \mid A, B)$$

Check it numerically. For each of the eight rows (a, b, c) : look up $P(A=a, B=b, C=c)$ straight from the table; then separately compute $P(A=a)$, $P(B=b \mid A=a)$ and $P(C=c \mid A=a, B=b)$ and multiply the three together. Print both numbers side by side and say whether they match (allow $1e-9$ for floating point).

Then answer in a comment: does the chain rule hold only for *this* table, or for every joint distribution? Why?

- T8. Bayes' rule.** You already know $P(A=1)$ — that was the worked example. Now suppose you are told that $C = 1$. Compute

$$P(A=1 \mid C=1)$$

and compare it with $P(A=1)$. Did learning $C=1$ make $A=1$ more likely or less likely? Write **one line** saying by how much, and in which direction.